Response to the Editor and Reviewers

Manuscript: *Spatio-temporal wall-actuation modes and cycle-mean viscous dissipation in a lid-driven cavity*

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Submission date: 21 November 2026

Point-by-point response. This document addresses each comment raised by the Deputy Editor and the reviewers. For every point, we reproduce the original comment in blue and provide our response and the corresponding revision in black. All changes are tracked in the marked-up manuscript (`manuscript_final_marked.pdf`) and the marked-up supplementary material (`supplementary_final_marked.pdf`).

Contents

General Statement

We thank the Deputy Editor and the reviewers for their constructive feedback. We have completely restructured the manuscript to place the **fluid physics** at the center of the contribution, as requested. The revised paper now addresses the following question:

> *How does the spatio-temporal structure of wall actuation determine the cycle-mean viscous dissipation of the resulting cavity flow at a prescribed actuation-energy level?*

The physics-informed neural network (PINN) is now presented as a tool for identifying actuation branches, not as the subject of the study. The independent multiple-relaxation-time lattice-Boltzmann (MRT-LBM) solver is used to evaluate the hydrodynamic response and the energetic cost of each branch. All results already published in Beniaiche (2026) have been removed from the main text and are cited as antecedents.

The revised manuscript contains **8 sections, 8 figures, and 4 tables**, well within the 15-figure limit. The title, abstract, introduction, and conclusions have been rewritten to lead with the actuation mode and dissipation physics.

Files submitted:

- `manuscript_final.tex` and `manuscript_final.pdf`: clean article file (source and author-generated PDF).
- `manuscript_final_marked.tex` and `manuscript_final_marked.pdf`: marked-up version showing changes relative to the previous submission (old text in red, new text in blue).
- `supplementary_final.tex` and `supplementary_final.pdf`: clean supplementary material.
- `supplementary_final_marked.tex` and `supplementary_final_marked.pdf`: marked-up supplementary.
- `RESPONSE_TO_EDITOR.tex` and `RESPONSE_TO_EDITOR.pdf`: this point-by-point response.
- `figures/highlight_image.png`: highlight image for the table of contents and potential cover.
- `figures/`: source files of the figures (the 8 embedded in the manuscript, the 2 in the supplementary, and the earlier diagnostic figures kept for reproducibility).

Response to the Deputy Editor

Editor Comment 1: "Physics of Fluids publishes original theoretical, computational, and experimental work that advances the physical understanding of fluids, meaning the mechanisms and principles that explain how and why fluids behave as they do. As written, the manuscript is framed as a reproducibility analysis of a physics-informed neural network controller for the lid-driven cavity. Before it can be considered, the paper needs to be reorganized so that the fluid physics is its central contribution."

Response: We have completely restructured the manuscript to place the fluid physics at the center. The new title, *Spatio-temporal wall-actuation modes and cycle-mean viscous dissipation in a lid-driven cavity*, leads with the actuation structure and the dissipation cost, not with the neural network.

The new section organization is:

1. **Introduction**: presents the physical problem of wall actuation in a driven cavity and the central question of cycle-mean dissipation at matched actuation norm.

2. **Physical formulation and energetic framework**: defines the quadratic actuation norm E_{act}, the viscous dissipation D, the mechanical wall power P_{lid}, and the energy budget $\overline{P}_{\mathrm{lid}} = \overline{D}$.

3. **Actuation modes and hydrodynamic states**: describes the stationary harmonic profiles and the three identified branches B1, B2, B3 as distinct hydrodynamic states, not as PINN outputs.

4. **Numerical methodology**: presents the PINN as a tool for identifying actuation branches and the MRT-LBM as the independent solver used to evaluate the flow response.

5. **Energetic response of stationary wall-actuation modes**: compares the dissipation of uniform, $k = 1$, $k = 2$, and B1 mean profiles at matched $E_{\mathrm{act}} = 0.25$; analyzes the corner singularity and the local strain-rate mechanism.

6. **Cycle-mean dissipation of time-dependent actuation branches**: demonstrates that $\overline{D(\mathbf{u})} \neq D(\overline{\mathbf{u}})$ and compares the cycle-mean dissipation of B1, B2, B3.

7. **Physical interpretation and implications for flow control**: explains why the second harmonic increases dissipation, why temporal forcing increases cycle-mean dissipation, and what branch accessibility tells us about physical performance.

8. **Conclusions**: summarizes the four main physical findings.

The PINN branch selection, regularization-dependent state recovery, and reproducibility audit have been moved to the supplementary material or cited as antecedents from Beniaiche (2026). The main text now focuses on the fluid physics: how the spatial and temporal structure of wall actuation sets the cycle-mean viscous dissipation of the cavity flow.

Editor Comment 2: "The physics to bring to the front is how the spatio-temporal structure of the wall actuation sets the cycle-mean viscous dissipation of the cavity flow. The neural network branch selection, the regularization dependent state recovery, and the reproducibility audit can remain as supporting evidence, but they should not frame the work."

Response: We have followed this recommendation exactly. The new abstract begins with the physical question:

> "The spatial and temporal organization of wall actuation determines how mechanical input is distributed across the flow structures of a driven cavity, yet the resulting

viscous-dissipation cost cannot be inferred from the actuation spectrum or from the time-mean velocity field alone."

The PINN is mentioned only as the method used to identify the actuation branches, which are then transferred to the independent MRT-LBM solver for evaluation. The reproducibility audit, the 185-realization campaign, the λ_{var} sensitivity study, and the seed multiplicity analysis have been moved to the supplementary material (Sections S1.1–S1.4, S2.3, S3.1–S3.3 and S5.1–S5.6).

Editor Comment 3: "The introduction and reference list should be repositioned toward the fluid physics literature on active control of the lid-driven cavity, viscous dissipation and energy budgets in actuated cavity flows, and modal structure and state selection in driven-cavity flow, so that the contribution is set against fluid flow."

Response: The new introduction (Sec. 1) begins with the physics of wall actuation in confined recirculating flows and cites the canonical cavity benchmarks [1, 2, 3, 4, 5] together with the flow-control literature carried over into the revised reference list (Gad-el-Hak 2000; Brunton & Noack 2015; Rowley & Dawson 2017; Bewley *et al.* 2001; Bergmann *et al.* 2005; Choi *et al.* 1994; Quadrio & Ricco 2004; Corke *et al.* 2010; Rabault *et al.* 2019; Garnier *et al.* 2021), followed by the dissipation and energy-budget literature (Pope 2000; Landau & Lifshitz; Doering & Gibbon; Beneddine *et al.* 2016). The reference list has been reorganized to prioritize fluid physics over machine learning. The preceding study [6] is discussed in one dedicated paragraph of the introduction, which clearly distinguishes the antecedent work (branch identification and numerical reproducibility) from the present contribution (cycle-mean dissipation and strain-rate mechanisms); it is cited elsewhere only where the present numbers are compared directly with that study.

Editor Comment 4: "Please also reduce the number of figures to maximum of 15."

Response: The revised manuscript contains **8 figures**, well below the 15-figure limit. The figures are:

1. Geometry and wall-actuation modes (new).
2. Spatio-temporal actuation signatures (new).
3. Grid convergence and reference sensitivity (old Fig. 10).
4. Spatial origin of dissipation differences (old Fig. 16, now central).
5. Iso-energetic comparison of the seven controls (old Fig. 11).
6. Flow topology of the actuation branches (old Fig. 13).
7. Mean velocity and fluctuation fields (old Fig. 14).
8. Phase-resolved mechanical-energy response (old Fig. 15).

Eleven figures of the previous version are no longer in the main text. Nine of them (old Figs. 1–9: branch-selection map, the 185-realization distribution in modal space, spectral fingerprints, the λ_{var} probability and entropy panels, the energy sweep, seed multiplicity, capacity/truncation and the Fourier vs. Chebyshev comparison) report the identification campaigns that are already published in Beniaiche (2026); they are now cited as antecedents, and the selection-frequency and sensitivity numbers that the present paper still refers to are tabulated in the supplement (Tables S2 and S4–S9). The remaining two (the quasi-stationarity diagnostic, old Fig. 12, and the symbolic-regression cross-validation, old Fig. 17) are not reported at all: neither is needed for the dissipation question posed here. The supplement contains two figures (Figs. S1–S2): the

eight phase bins of the branch captures that underlie Main Figs. 6–8, and the same layout with a separate stream-function scale per panel.

Editor Comment 5: "Please revise the title to lead with the actuation mode and dissipation physics."

Response: The new title is:

Spatio-temporal wall-actuation modes and cycle-mean viscous dissipation in a lid-driven cavity

This title directly addresses the editor's request: it leads with the actuation structure ("spatio-temporal wall-actuation modes") and the dissipation physics ("cycle-mean viscous dissipation"), and it situates the problem in the lid-driven cavity.

Editor Comment 6: "Please revise your manuscript and submit a detailed, point-by-point response to the comments. Each response should include the original comment followed by a description of the corresponding revision made (or a justification if no change was made). The revised manuscript and response are due by 21-Nov-2026 via the PXP web site."

Response: This document is the point-by-point response. Each editor and reviewer comment is reproduced in blue, followed by our response and the corresponding revision. The revised manuscript is due by 21 November 2026.

Editor Comment 7: "When submitting your revision, please upload two versions of your manuscript: a clean version labeled 'Article File' and a marked-up version labeled 'Marked-Up Manuscript,' which should clearly show all changes made from the previous submission."

Response: We have prepared both versions:

- **Clean version**: `manuscript_final.tex` and `manuscript_final.pdf` (26 pages).
- **Marked-up version**: `manuscript_final_marked.tex` and `manuscript_final_marked.pdf` (34 pages). The marked-up version uses a paragraph-level diff: old text that has been removed or replaced is shown in red, new or rewritten text is shown in blue. Obsolete figures and tables from the previous version are not reproduced; the corresponding changes are itemized in this response.

The same approach has been applied to the supplementary material:

- **Clean version**: `supplementary_final.tex` and `supplementary_final.pdf` (27 pages).
- **Marked-up version**: `supplementary_final_marked.tex` and `supplementary_final_marked.pdf` (37 pages).

Editor Comment 8: "Revisions must be submitted in the proper file formats. When submitting your revision, please upload the source file for your revised manuscript. Figures must be embedded in the text and visible in the PDF. An author-generated PDF must also be provided since individual figure files are not being uploaded."

Response: We have uploaded:

- **Source files**: clean sources `manuscript_final.tex`, `supplementary_final.tex` (with its `data_appendix/` table files), marked-up sources `manuscript_final_marked.tex` and `supplementary_final_marked.tex`;
  this letter, `RESPONSE_TO_EDITOR.tex`.

- **Author-generated PDFs**: `manuscript_final.pdf` (26 pages), `supplementary_final.pdf` (27 pages), the two marked-up PDFs (34 and 37 pages), and `RESPONSE_TO_EDITOR.pdf`.
- **Figures**: the 8 figures of the manuscript and the 2 of the supplementary material are embedded in the two PDFs and visible in the text. Their source files (PDF and PNG) are provided in the `figures/` directory.

Editor Comment 9: "You will also be asked to upload a highlight image that will be used in the table of contents if your manuscript is published. The highlight image should be a visually interesting figure representing your manuscript and may also be selected for the Cover Image."

Response: We have prepared `highlight_image.png` (2220×900 px at 300 dpi). It shows, for the three identified actuation branches B_1 (quasi-steady), B_2 (lid-oscillatory) and B_3 (mixed) at matched actuation norm $\overline{E}_{\mathrm{act}} = 0.25$ and $Re = 500$, the spatial distribution of the cycle-mean local dissipation density $\langle \phi \rangle$ reconstructed from the eight phase bins of the captured periodic solution, with the streamlines of the time-mean flow overlaid and the cycle-mean dissipation $\overline{D}$ annotated on each panel. The three panels use a logarithmic colour scale of five decades, so the image compares *where* the dissipation is produced rather than its absolute amplitude. This is exactly the central finding of the paper: at matched actuation energy the three branches organise the same cavity into radically different dissipation patterns — space-filling two-cell dissipation for the quasi-steady branch B_1, dissipation confined to the oscillating Stokes layer under the lid for B_2 (whose time-mean flow carries only 25% of the cycle-mean kinetic energy), and an intermediate organisation for B_3 — and therefore different cycle-mean viscous costs. The image is generated from the archived field captures by `scripts/make_highlight_image.py`, so it is reproducible and uses no data other than those reported in the manuscript. It is visually striking (three contrasting topologies on one theme) and suitable for the table of contents and potential cover.

Response to the Reviewers

Note: The reviewers' comments are not reproduced here because the Deputy Editor's comments already captured the main concerns. Points 1–14 below follow the revision plan derived from those comments; Points 15–16 additionally document the temporal mapping of the exported actuation onto the lattice, which is the protocol item a reader of Sec. 4 needs in full; Points 17–21 record the changes made after a further audit of the identification code and of the discrete wall-power estimator, which is the protocol item a reader of Secs. 2.2 and 4.1 needs in full; and Points 22–25 record the last pass over the written time-step formula, the two lattice spacings the implementation actually uses, the status of the identification's space–time field, and the comparison quoted in the abstract.

Point 1: New title, abstract, and introduction.

The new title is *Spatio-temporal wall-actuation modes and cycle-mean viscous dissipation in a lid-driven cavity*. The new abstract begins with the physical question and announces the main quantitative results (17.7% reduction for $k = 1$, 36.8% and 44.3% increase for $k = 2$ and B1 mean, fluctuating shares of 92.9%, 53.7% and 1.8% of the cycle-mean cost for B2, B1 and B3, and a cycle-mean spread of the time-dependent branches by the modest factor of 1.2). The new introduction (Sec. 1) presents the physics of wall actuation, the energy budget, and the central question.

Point 2: Reorganization of sections around the energy budget.

The 8 sections follow the physical storyline: problem formulation → actuation modes → numerical methodology → stationary results → time-dependent results → physical interpretation → conclusions. The energy budget $\overline{P}_{\mathrm{lid}} = \overline{D}$ is introduced in Sec. 2 and used throughout.

Point 3: Elimination of repetitions with the published article.

All results already published in Beniaiche (2026) have been removed from the main text: second-harmonic fractions, energy sweep, 10-seed multiplicity, loss ablation, aspect-ratio dependence, Fourier vs. Chebyshev, quasi-stationarity diagnostic, same-lid-speed dissipation comparison, and symbolic regression. The campaigns that the present argument still refers to are now reported as supplementary tables (Tables S2 and S4–S9) and cited as antecedents; the quasi-stationarity diagnostic and the symbolic-regression study, which the dissipation question posed here does not use, have been dropped from both the manuscript and the supplement.

Point 4: Conservation of iso-energetic comparisons and strain-rate mechanisms.

The stationary comparison at matched $E_{\mathrm{act}} = 0.25$ (Table 2) and the local strain-rate analysis (Fig. 4) are now central results. The corner singularity and the reference dependence (Sec. 5.2) are preserved.

Point 5: Improvement of the power–dissipation budget.

The energy budget (Table 4) is presented as a verification with wall-shear estimate uncertainty, not as an exact closure. The first-order estimator gives residuals of -2.8%, -0.6% and -0.4% for B1, B2 and B3. Replacing it by a second-order one-sided derivative moves the B1 *mean-field* residual from -3.06% to $+0.39\%$, i.e. across zero, which bounds the estimator uncertainty at a few per cent, and refining the lattice from $N = 256$ to $N = 512$ divides the three residuals by 2.1, 1.9 and 2.4 (see supplementary Section S4.8 and Points 15–16 below). We do not claim exact closure but document the uncertainty.

Point 6: Reorganization of figures and tables.

The 8 figures and 4 tables are listed in the response to Editor Comment 4. The figure order follows the physical storyline: geometry → signatures → convergence → mechanism → iso-energetic comparison of the seven controls → topology → mean and fluctuation fields → phase-resolved budget.

Point 7: Transfer of complementary PINN campaigns to the supplementary.

The 185-realization campaign, the λ_{var} sensitivity study, the seed multiplicity analysis, and the PINN hyperparameter tables have been moved to the supplementary material (Sections S1.1–S1.4, S2.3 and S3.1–S3.3; the reproducibility inventory is in Sections S5.1–S5.6).

Additionally, the energy sweep campaign (Section S5.1 and Table S5 of the version originally submitted) has been **removed entirely** from the supplementary material. This analysis was already documented in our previous publication (Beniaiche 2026) and is not required for the physics-first narrative of the present manuscript. The new work focuses on the representatives B1, B2, and B3 at $Re = 500$, $E^* = 0.25$, whose characterization does not depend on the energy sweep.

Point 8: Update of the bibliography and citation of Beniaiche (2026).

The bibliography contains 37 references, prioritizing fluid physics over machine learning. Beniaiche (2026) is cited in the introduction and in the bibliography as:

> A. Beniaiche, "Physics-informed identification and independent numerical verification of conditional modal collapse in active control of lid-driven cavity flow," *Phys. Fluids* **38**, 103612 (2026).

Point 9: Rewriting of the conclusions and response to the Deputy Editor.

The new conclusions (Sec. 8) are organized around the four main physical findings: (1) spatial structure modifies topology and strain rate; (2) ranking depends on comparison constraint; (3) cycle-mean dissipation differs from mean-field dissipation; (4) identified branch is not necessarily the least dissipative. The PINN is mentioned only as a tool for identifying states.

Point 10: Explicit distinction between E_{act} and P_{lid}.

Section 2.2 clearly states that $E_{\text{act}} = \int U_{\text{lid}}^2 \, dx$ is a quadratic actuation norm, not the mechanical work transmitted to the fluid. The mechanical wall power is $P_{\text{lid}} = \int U_{\text{lid}} \tau_w \, dx$. On cycle average, $\overline{P}_{\text{lid}} = \overline{D}$.

Point 11: Addition of $U_{\max}/Re_{\max}$ column to the stationary table.

Table 2 now includes a column for $U_{\max}$ and $Re_{\max} = U_{\max} L/\nu$. At $E^* = 0.25$, the uniform profile has $U_{\max} = 0.50$ ($Re_{\max} = 250$), while the harmonics have $U_{\max} = 0.7071$ ($Re_{\max} = 354$). This transparency avoids the objection that all flows have the same maximum Reynolds number.

Point 12: Energy budget presented as verification with uncertainty.

Section 6.5 and Table 4 present the energy budget as a verification with wall-shear estimate uncertainty. The recovered-velocity, first-order estimator gives cycle-mean residuals $\overline{P}_{\text{lid}}^{\text{rec}} - \overline{D}$ of -2.8%, -0.6% and -0.4% for B_1, B_2 and B_3, i.e. the budget closes to a few per cent. The estimator built on the *prescribed* lid velocity gives $+3.4$, $+4.6$ and $+4.7\%$, and the two brackets bound the estimator uncertainty (Section S4.8). We do not claim exact closure. All of these percentages, together with the alternative estimators, are recomputed from the archived field captures by `scripts/d2_derived_quantities.py`, whose output is `results/d2_derived/d2_summary.json`; the residual annotated in Fig. 8 is the one of the two recorded signals (-2.8, -0.6 and -0.4%), and the caption states which estimator each number comes from.

Point 13: Marked-up versions built by sharing the new preamble.

The marked-up manuscript and supplementary were built using a paragraph-level diff script (`mark_manuscript.py`) that shares the new preamble for both the old and new bodies. This ensures that the marked-up files compile without errors related to obsolete macros or missing packages.

Point 14: Deep point-by-point verification.

We have performed a deep verification of the revision plan and the editor's comments. The reproducibility inventory and the data-availability statement are provided in the supplementary material (Sections S5.5–S5.6). All points have been addressed.

Point 15: Temporal mapping of the exported actuation onto the lattice.

The protocol is stated in Sec. 4.2 of the manuscript and verified in supplementary Sections S4.3 and S4.5. The actuation is defined over the physical nondimensional period $T = 2$, with Fourier components $\cos(2n\pi t/T) = \cos(n\pi t)$, so the fundamental actuation frequency is $\omega_1 = 2\pi/T = \pi$ in cavity units on every grid. The lattice advances by $\Delta t = U_{\mathrm{ref}}\Delta x = U_{\mathrm{ref}}/N$ with $\Delta x = 1/N$ and $U_{\mathrm{ref}} = 0.05$; one physical period therefore spans $N_T = TN/U_{\mathrm{ref}} = 40N$ lattice steps, i.e. $N_T = 10\,240$ at $N = 256$ and $20\,480$ at $N = 512$ (Eq. 14 of the manuscript). Each of the 256 exported samples of the waveform is held for $N_T/256$ consecutive steps, i.e. 40 at $N = 256$ and 80 at $N = 512$, so its physical duration $T/256 = 7.8125 \times 10^{-3}$ is resolution-independent and the staircase it imposes is common to both grids. The largest instantaneous lid speed of the campaign is $1.23\,U_{\mathrm{ref}}$, hence Ma ≈ 0.11, and the oscillatory wall-layer thickness $\delta_1 = \sqrt{2\nu/\omega_1} = 0.036$ is about nine cells at $N = 256$ and eighteen at $N = 512$, so the same margin resolves that layer at both resolutions.

Three checks close the mapping. *(i) Frequency response.* A dedicated run whose lid is the pure harmonic $\sin(2\pi x)\cos(\omega_1 t)$, at the same E^*, Re, N and ω_1 (`scripts/validate_stokes_analytic.py`, Section S4.3) records at the antinode a single-sided amplitude of 0.927 at ω_1 and no other significant line — the imposed unit amplitude, recovered to within the 7.3% slip of the velocity boundary condition — and a wall-normal exponential decay of fitted length 5.4 cells ($R^2 = 0.999$ over the eleven cells nearest the wall) against the half-space value $\delta = \sqrt{2\nu/\omega_1} = 9.1$ cells. The layer is of the expected order and 41% thinner, which is attributed to the non-uniform lid, the finite depth (about 28δ) and the steady shear superimposed by the return flow; the run is used as a frequency-response check of the solver and of the mapping, not as a quantitative benchmark of the layer thickness. *(ii) Spectral purity of the branch runs.* A streamwise probe at the cavity centre, recorded over the whole sampling window, has its spectral peak at 3.141593 for all three branches, with 97.5–99.5% of its fluctuation energy in the five spectral bins around that peak, and the dissipation signal peaks at ω_1 for the quasi-stationary B_1 and at $2\omega_1$ for B_2 and B_3 (Section S4.5). *(iii) Effect of the hold.* Replaying B_2 at $N = 256$ with the exact sample-time command instead of the 40-step hold returns $\overline{D} = 1.4425 \times 10^{-2}$ against 1.4408×10^{-2}, i.e. +0.114%, with the fluctuating part moved by +0.12% and the kinetic energy by +0.005%, while the realized cycle-mean actuation energy stays within 1×10^{-16} of the target $E^* = 0.25$ on both grids; the staircase is therefore not the source of the reported differences.

Every cycle mean quoted in the paper is a direct time average over the 32 retained forced periods, after a transient terminated by a drift test (24 forced periods, successive three-period windows changing by less than 2×10^{-3}). The eight-bin phase quadrature of the same data reproduces $\overline{D}$ to within 0.004% but under-estimates the peakier lid power by 0.14%, 5.5% and 3.9% for B_1, B_2 and B_3, so the quadrature is used only as a cross-check.

Point 16: Grid-refinement sensitivity of the time-dependent comparison.

The three branch runs were replicated at $N = 512$ under the same temporal protocol, and the comparison is reported as a *grid-refinement sensitivity*: two grids bound the variation of the cycle means, they do not determine a convergence order, so no extrapolation and no convergence index is computed for them (Sec. 4.3 and Table S15). The refined runs raise $\overline{D}$ by 5.4%, 7.3% and 7.1% for B_1, B_2 and B_3, preserve the ordering $B_2 > B_1 \simeq B_3$, and narrow the B_1–B_3 gap from 2.1% to 0.5%, so the central conclusion of Sec. 6 — that the three branches lie within a factor 1.23 of one another on the cycle average while their fluctuating parts span two orders of magnitude — is unchanged by the refinement. The power-balance residual of Sec. 6.5 is divided by 2.1, 1.9 and 2.4, i.e. the factor of two a first-order one-sided stencil predicts under $\Delta x \to \Delta x/2$, and moves to −1.4%, −0.3% and −0.2%, while the largest slip of the velocity boundary condition drops by 38–47%. The periodicity of the averaged state is not degraded: the half-window phase reproduction of the finer captures is 1.79×10^{-3}, 1.48×10^{-5} and 2.69×10^{-4} against 1.87×10^{-3}, 1.46×10^{-5} and 2.61×10^{-4} at $N = 256$. Each finer run cost ≈ 13.1 h of single-threaded wall time against ≈ 0.7 h for the corresponding $N = 256$ capture, which is why the phase-resolved fields

are reported on the $N = 256$ realization and the cycle-mean scalars on both grids. Sec. 4.3 and Supplementary Sec. S4.5 now state that division of labour explicitly — the replication supplies scalar diagnostics only, and every field and phase-resolved quantity in the paper is an $N = 256$ capture — so that no figure is read as finer than it is.

Point 17: The training window and the export window of the identification, now distinguished.

A fresh audit of the training script confirms that the identification samples the nondimensional time τ uniformly over $[0, T_{\max}]$ with $T_{\max} = 1$, that the control is expanded there as $\sin(m\pi x)\cos(n\pi\tau)$, and that the momentum residual differentiates with respect to τ directly, with no $1/T$ chain factor. The manuscript and the supplement previously described that same variable as the normalized phase $\theta = t_{\text{phys}}/T$ of the exported period, which conflated the training window with the export window. Sec. 4.1 and Supplementary Secs. S1.1, S1.2 and S1.4 now state the distinction explicitly, in the order the computation performs it: the coefficient network returns the amplitudes $a_{mn}(Re)$, which carry no time variable; the flow network is trained on $\tau \in [0, 1]$, which covers one half-period of the fundamental; and the exported wall actuation is the *same* Fourier series re-evaluated on the same nondimensional time over its complete period, $\cos(2n\pi t_{\text{phys}}/T) = \cos(n\pi t_{\text{phys}})$ with $0 \le t_{\text{phys}} \le T = 2$, so the fundamental of what the lattice actually sees is $\omega_1 = 2\pi/T = \pi$ (Sec. 4.2, Eq. 14, Supplementary Sec. S2.1). The PINN flow field is not periodicised and not transferred to the solver — no endpoint condition is imposed on it — and the periodic hydrodynamic response is established independently by the MRT-LBM calculation and verified a posteriori to 10^{-3} in relative field norm (Supplementary Sec. S1.4, and see Point 24 below for the status now given to that field). We also state why the constraint is not affected by the shorter training window: $\int_0^1 \cos^2(n\pi\tau)\, d\tau = \int_0^1 \cos^2(2n\pi\theta)\, d\theta = 1/2$ with vanishing cross-terms on both windows, so the penalised quantity is the cycle mean of the exported signal even though the flow field was trained on half a period (Supplementary Sec. S2.2).

Point 18: Direction of the Stokes comparison.

The abstract, the contribution list of Sec. 1, Sec. 6.3 and Sec. 8 previously said that the second-Stokes estimate "accounts for 95% and 98%" of the fluctuating dissipation. That is the reciprocal of what the numbers say: the measured fluctuating parts are 1.339×10^{-2} for B_2 and 6.278×10^{-3} for B_3 against estimates of 1.411×10^{-2} and 6.383×10^{-3}, so the measurements are 95% and 98% *of the estimate* and the estimate is slightly the larger. The four places now read that "the estimate agrees with the measured fluctuating dissipation to within approximately 5% for B_2 and 2% for B_3, without an adjustable constant", which states the accuracy of the approximation rather than a share of the budget. For the same reason the abstract no longer claims the estimate "locates" the cost in the wall layer: the model is a half-space one, so the agreement supports a wall-layer interpretation of the dominant fluctuating cost rather than a spatial decomposition of the total, and Sec. 7.1 adds explicitly that the 56% the estimate leaves unexplained for the $k = 1$ profile is not thereby assigned to the cavity core.

Point 19: Definition of R_K.

R_K was described in the nomenclature as the "ratio of fluctuating to mean kinetic energy", but the reported values are computed against the total. With $\overline{K} = \overline{K}_{\text{mean}} + \overline{K}_{\text{fluct}}$ the definition is now $R_K = \overline{K}_{\text{fluct}}/\overline{K}$, "the fraction of the cycle-mean kinetic energy carried by temporal fluctuations", which is what reproduces the table: B_1 0.16% against a mean field holding 99.8%, B_2 74.9% against 25%, B_3 12.5% against 87.5%. The two supplement statements that read "fluctuates by R_K of its mean" in Secs. S4.3 and S4.5 now read "of the cycle-mean kinetic energy", which matters most for B_2, where the old wording would have implied that the mean field retains three quarters of the energy instead of a quarter.

Point 20: Geometry of the discrete wall-power estimator.

The main text and the supplement disagreed about where $\overline{P}_{\text{lid}}^{\text{rec}}$ is evaluated: Sec. 2.2 placed

both factors of the product on the plane $y = 1 - \Delta y$ (index $N - 2$) while Supplementary Sec. S4.8 wrote them on row $N - 1$. We re-read the estimator in the solver, and it uses the lid node row for both factors, $U_{\mathrm{wall}}^{\mathrm{rec}}(x_j) = u_x(N-1, j)$ and $\tau_w^{\mathrm{FD}}(x_j) = Re^{-1}[u_x(N-1, j) - u_x(N-2, j)]/\Delta y$, with trapezoidal end weights. Eq. (5) now carries those discrete definitions and Sec. 2.2, the caption of Table 4, Sec. 6.5 and Supplementary Sec. S4.8 use the same single convention. The accompanying statement about the boundary condition was corrected in the same pass: the lid is imposed by half-way bounce-back with momentum injection into the two wall-facing populations, which enforces the moving wall at the hydrodynamic boundary half a lattice spacing outside row $N - 1$; it therefore does *not* make the recovered node velocity equal the commanded profile, and Supplementary Sec. S4.1 no longer claims it does. That discrepancy is the slip already quantified in Sec. 6.5 — 6.0%, 11.5% and 11.8% of the cavity speed at $N = 256$, dropping to 3.2%, 7.2% and 6.9% at $N = 512$ — and Sec. 2.2 now explains the residuals of Table 4 through it, instead of leaving the reader to infer why the recovered and prescribed estimators bracket the closure.

Point 21: Quantitative statements made exact.

Four imprecise phrases were replaced by the numbers they summarise. Sec. 6.2, the caption of Fig. 5 and Sec. 8 said the branch totals lie "within twenty per cent" of one another; B_2 is in fact 20.7% above B_1 and 23.2% above B_3 at $N = 256$, and Sec. 6.2 and the Conclusions now give both ratios and bound the spread by a factor 1.23. "Indistinguishable within 2%" for B_1 and B_3 became "nearly equal, 2.1% apart at $N = 256$ and 0.5% apart at $N = 512$" in the abstract, Sec. 6.2, Sec. 8 and Supplementary Sec. S4.5, which keeps the claim true at both resolutions instead of rounding one of them. The nomenclature now distinguishes U_0, the scaling velocity of the nondimensional problem, from $U_{\mathrm{ref}} = 0.05$, the lattice reference velocity that sets $\Delta t = U_{\mathrm{ref}}/N$, and lists E_{act} as dimensionless with dimensional equivalent $U_0^2 L$, since the constraint is written in the scaled variables of Eq. (1). Finally, the single-harmonic test at $n = 1, 2, 4$ is stated as work not yet done in Sec. 6.3 and Sec. 8: Supplementary Sec. S4.5 tabulates the harmonic budgets of the three controls already used, it does not contain a mono-harmonic campaign, and the sentence pointing to it no longer suggests otherwise.

Point 22: The written formula for the lattice time step.

Supplementary Sec. S4.5 stated $\Delta t = \Delta x/U_{\mathrm{ref}}$ in the same paragraph that announces $N_T = TN/U_{\mathrm{ref}} = 40N$, and Main Sec. 4.2 states $\Delta t = U_{\mathrm{ref}}/N$. The reviewer is right that the two are not equivalent, and right that the second is the mapping the code performs: in `protocol()` of `scripts/run_branch_capture_d2.py` the timestep is `dt = U_ref/N` and the number of steps per period is `T*N/U_ref`, and the function aborts the run if the product of the two differs from T by more than 10^{-12}. Sec. S4.5 now reads $\Delta t = U_{\mathrm{ref}}\Delta x = U_{\mathrm{ref}}/N$, and it adds the viscosity identity that only this convention satisfies: with $\Delta x = 1/N$ and $\Delta t = U_{\mathrm{ref}}/N$, the lattice viscosity $\nu_{\mathrm{latt}} = U_{\mathrm{ref}}N/Re = 0.0256$ at $N = 256$ converts to $\nu_{\mathrm{phys}} = \nu_{\mathrm{latt}}\Delta x^2/\Delta t = 2 \times 10^{-3} = 1/Re$ exactly, which is the viscosity the physical problem is posed with. The three consequences the reviewer lists — $N_T = 10\,240$ and $20\,480$, the measured $\omega_1 = \pi$, and ν_{latt} — are therefore all consequences of the single written mapping, and the paragraph in Sec. S4.5 now says so. Point 15 of this letter, which reproduced the mistyped formula, was corrected in the same pass. No computed quantity changes: the campaign was driven by the code, and the error was in the text.

Point 23: The two spacings, resolved against the executed code.

As instructed, this was settled by reading the implementation rather than by harmonising the prose, and the finding is that the implementation uses both spacings, at two different levels. The solver's spacing is $\Delta x = 1/N$: the lattice holds N fluid node rows per side, the half-way boundary condition places each wall half a spacing beyond the row it faces (Sec. S4.1), so a unit cavity spans $N\Delta x$ and this is the spacing behind Δt, N_T, ν_{latt} and the cell count of δ_1. The post-processing works on the archived fields, whose coordinate arrays are $x_j = y_j = j/(N-1)$: the derived quantities and the pointwise diagnostics therefore pair the first and last node with the walls

and differentiate with $h = 1/(N-1)$ (`d2_derived_quantities.py`, `d2_topology_numbers.py`, `run_branch_capture_d2.py`). The two spacings differ by 0.39% at $N = 256$ and 0.20% at $N = 512$, and the diagnostic layout also absorbs the half-cell wall offset into that same $O(1/N)$ convention.

What that difference does to the reported numbers was then measured instead of argued. The new `scripts/spacing_convention_check.py` rebuilds, from the same archived fields, the dissipation of the mean field, the dissipation of the fluctuating field, the eight-bin phase reconstruction of $\overline{D}$ and the recovered lid power, once with $h = 1/(N-1)$ and once with $\Delta x = 1/N$, at both resolutions and for all three branches. Every one of them changes by less than 5×10^{-16} in relative terms, because a squared derivative (h^{-2}) is always paired with an area element (h^2) in the discrete dissipation, and the first-order wall shear (h^{-1}) is paired with a line element (h) in the lid power. The convention survives only where a measure is not paired with a derivative: the kinetic energies scale as h^2, so they are 0.78% ($N = 256$) and 0.39% ($N = 512$) smaller under $\Delta x = 1/N$, and a pointwise derivative such as the steady-streaming forcing scales as h^{-1}, i.e. 0.39% and 0.20% larger. Both are single multiplicative factors, identical for the three branches and for the reference lids, so they cannot alter a branch-to-branch ratio, a ranking, the 2.1%/0.5% B_1–B_3 comparison, or any figure reading; the largest of them is more than an order of magnitude below the 5.4%–7.3% refinement variation that the sensitivity study reports. A quotient of two quantities of *different* differential order is sensitive in the same way, and this had been missed in the first version of the paragraph: the forcing-to-viscous-term ratio of Sec. S4.7 pairs one derivative with two, so it scales as h and shrinks by the same 0.39%/0.20%. Point 26 below reports the measurement of that case and the text that replaces the withdrawn claim.

The text now says which spacing enters where instead of implying one convention throughout: Sec. S4.5 gained a subsubsection "Which spacing enters where ($\Delta x = 1/N$ and $h = 1/(N-1)$)" with the measurements above and the artifact `results/d2_derived/spacing_convention_check.json`, Main Sec. 4.2 carries the short version, Sec. S4.7 now names its h as the node spacing of the archived arrays rather than "the physical spacing", with the correction of Point 26 for the ratio it forms with the viscous term, and the discrete estimator of Sec. S4.8 is written with h — including its derivative — with the statement that rebuilding it with $\Delta x = 1/N$ leaves $\overline{P}_{\mathrm{lid}}^{\mathrm{rec}}$ unchanged to machine precision. Sec. 2.2 of the main text defines Δy as the node spacing of those arrays, so Eq. (5) and Sec. S4.8 now use one symbol with one meaning. No reported value was changed by this point.

Point 24: Status of the PINN space–time field.

The claim is withdrawn. Sec. S1.4 previously said that, since the identification solves a space–time boundary-value problem with no initial-condition term, "training targets a stationary response of that problem rather than the transient from a prescribed initial field". The reviewer is correct that this does not follow: the training imposes neither an initial field nor a periodicity condition on the flow network, so for the instationary system the losses do not select a unique hydrodynamic trajectory, and the shorter training window $[0, 1]$ does not repair that. S1.4 now describes the space–time field as what it is in the computation — an auxiliary field in the joint minimisation of the PDE, boundary and control losses over $0 \le \tau \le 1$, explicitly not interpreted as a unique transient solution nor as the periodic hydrodynamic response to the exported actuation — and states that the only objects transferred are the coefficients a_{mn}, the periodic response used by every dissipation number being computed independently by the MRT-LBM solver and verified a posteriori (half-window phase reproduction below 1.9×10^{-3} in relative field norm, Sec. S4.6). This matches what the paper actually does and does not overstate the identification.

Sec. S1.2 was reworded in the same direction, along the lines the reviewer proposed: τ is the same nondimensional time as the one appearing in the Navier–Stokes equations of Main Sec. 2.1, and the training merely restricts it to the interval $[0, 1]$; export re-evaluates the same analytic Fourier series over the complete period, $t_{\mathrm{phys}} \in [0, 2]$. The words "raw training clock" and "a different clock" were removed from S1.2 and S4.5, and the two remaining places that described

τ as a "raw nondimensional training time" or "raw training variable" — the architecture list of Sec. S1.1 and the sampling protocol of Sec. S1.3 — now call it the nondimensional time of the governing equations restricted to the training window, so that the text no longer suggests two physical time scales, while the distinction that does matter — training window versus export window, and the absence of a $1/T$ chain factor in the residual — is kept stated in the order the code performs it.

Point 25: The abstract's actuation-energy comparison.

The abstract opposed two actuations "whose oscillatory energies differ by a factor of 2.3"; that is the B_2–B_3 pair, whose dissipation differs by 21%–23%, so the sentence illustrated nothing that the paper claims. The comparison the paper rests on is B_1 against B_3: at the same $E^* = 0.25$ their oscillatory actuation energies are 3.13×10^{-3} and 0.1120, a factor of 35.7, while their cycle-mean dissipations are 1.1934×10^{-2} and 1.1690×10^{-2}, a difference of 2.1% at $N = 256$ and 0.5% at $N = 512$. The abstract now states exactly that, including the mechanism — the two totals are reached through opposite partitions between mean field and fluctuations — and Sec. 6.2 was changed from "two branches whose totals agree to within 2%" to "differ by only about 2.1%", so abstract, body and tables now quote the same precision. Both numbers are recomputed from `results/d2_derived/d2_summary.json` and `results/d2_derived/resolution_sensitivity_hold.json`, and the $\overline{D}$ ordering $B_2 > B_1 \simeq B_3$ is preserved at both resolutions.

Point 26: The forcing-to-viscous-term ratio does carry the spacing factor.

The reviewer is right and the sentence was wrong. Sec. S4.7 claimed that the h^{-1} factor of the pointwise Reynolds-stress forcing "is identical for the three branches and for the viscous term it is compared with, so it cancels in every ratio quoted below." The two terms are not of the same differential order: $\mathbf{F}$ is one divergence of the stress, so it scales as h^{-1}, while $\nu\nabla^2\overline{\mathbf{u}}$ carries two derivatives and scales as h^{-2}. Their quotient therefore scales as h, and passing from $h = 1/(N-1)$ to $\Delta x = 1/N$ multiplies it by $(N-1)/N$, a reduction of 0.39% at $N = 256$ and 0.20% at $N = 512$ — exactly the numbers the reviewer computes.

This was again measured rather than argued. `scripts/spacing_convention_check.py` now rebuilds the ratio of Sec. S4.7 under both spacings with the identical stencils of `d2_topology_numbers.py` (central differences, the same three-node exclusion, phase-mean stress and mean-field Laplacian): it reproduces the published rms ratios 0.259, 0.067 and 0.0027 for B_2, B_3 and B_1 and returns $(N-1)/N$ for the change in every branch, to the last digit (new keys `forcing_over_viscous_ratio_` of `results/d2_derived/spacing_convention_check.json`). Sec. S4.7 states the h-scaling and those two percentages, and says explicitly that the quoted ratios are the $h = 1/(N-1)$ values; the inventory in Sec. S4.5 gained a quotient-of-different-orders entry next to the kinetic energies and the pointwise forcing; and Main Sec. 4.2 lists the same factor. The blanket sentence "it alters no ratio" is now restricted to what it was always true of: being multiplicative and common to the three branches, the spacing alters no branch-to-branch ratio, no ranking, no B_1–B_3 comparison and no figure reading. Table S17 keeps the $h = 1/(N-1)$ values it was computed with, so no reported number changed; only the claim that the spacing cancels in that quotient was withdrawn.

Point 27: Distinct modal families, not coexisting hydrodynamic states.

The introduction previously opened the branch discussion with "A periodic boundary-value problem of this type admits several coexisting branches," which the experiments do not support: the three controls are three different wall signals, each evaluated in its own run, so they show that the identification recovers several families, not that one prescribed lid motion has several periodic responses. The paragraph now reads, in the reviewer's formulation, that spatially structured and time-dependent wall actuations can be recovered in distinct modal families, that the family selected depends on the formulation, the regularization and the initialization of the optimization,

that these families are branches of the *control-identification landscape* and not evidence of several hydrodynamic states coexisting under one and the same prescribed lid motion, and that each exported actuation is treated here as a separate boundary forcing whose periodic response is computed independently with the MRT-LBM solver. A sweep of both documents confirms that no other passage claims coexistence, bistability or hysteresis of hydrodynamic states: the only remaining occurrence of the word is the sentence that denies it.

Point 28: The periodicity bound of the LBM fields.

Sec. S1.4 announced half-window discrepancies "below 1.0×10^{-3} for all three branches" while Secs. S4.2, S4.5 and S4.6 report 1.9×10^{-3} (B_1), 2.6×10^{-4} (B_3) and 1.5×10^{-5} (B_2) at $N = 256$, and 1.79×10^{-3}, 2.69×10^{-4} and 1.48×10^{-5} at $N = 512$. The bound was the part that was wrong, not the measurements. Sec. S1.4 now lists the three values at $N = 256$, adds the $N = 512$ triple, states the global bound 2×10^{-3} — better than 0.2% reproduction of the phase-averaged field between the two halves of the window — and points to Secs. S4.2 and S4.6 for the check itself instead of to Sec. S4.5, which is where the finer runs are reported; the forward pointer in the same subsection was corrected in the same way. The caption of Table 4 of the main text, which quoted the same 10^{-3} bound, now gives the three measured values as well. No measured quantity changed.

Point 29: Which denominator the B_1 Stokes comparison uses.

"16%" in the main text and "19%" in the supplement were the same gap expressed against two different denominators, and neither document said which. From the published numbers, $1 - 1.8143 \times 10^{-4}/2.1491 \times 10^{-4} = 15.6\%$ (the estimate below the measurement) and $2.1491 \times 10^{-4}/1.8143 \times 10^{-4} - 1 = 18.5\%$ (the measurement above the estimate, which is the 118% ratio column of Table S16). Sec. S4.5 now states both, names 18.5% as the convention of that ratio column, and identifies the main text's 16% as the shortfall of the estimate relative to the measurement; Main Sec. 6.3 is kept as the reviewer preferred. The sentence also now says what the 1.8% refers to: the increment itself is only 1.8% of the B_1 cycle mean, which is why the comparison is not over-read.

Point 30: The wall is a half-way plane, not a lattice node.

The expression "enforces that velocity at the half-way wall node", left over in Sec. S4.8, did contradict Sec. S4.1. Sec. S4.8 now reads, as proposed, that the boundary condition imposes the commanded moving-wall velocity at the hydrodynamic boundary — the half-way plane a half spacing outside the lid node row — and that the macroscopic streamwise velocity reconstructed at the adjacent outermost fluid-node row $i = N - 1$ is a computed quantity that differs from the prescribed profile by the discrete slip of the scheme. The same sweep renamed the remaining loose uses: the list of estimator choices in Sec. S4.8 speaks of the velocity at the lid node row, the sentence after Eq. (5) of the main text no longer says the boundary condition fails "at the wall node" but that it imposes the velocity at the hydrodynamic wall outside that row, and the captions of Table 4 and Fig. 5 now say the recovered velocity is the one of the outermost fluid row rather than "the boundary node". The slip magnitudes themselves (0.060, 0.115, 0.118 cavity units) are unchanged.

Point 31: What the identification actually solves.

As proposed, Sec. S1.4 now calls the problem what the loss functions define it to be: a space–time PDE-constrained optimization problem on the training slab $(x, y, \tau) \in [0, L_x] \times [0, 1] \times [0, 1]$, in place of "space–time boundary-value problem in the training variables". The slab, the basis and the absence of any initial or periodicity constraint are stated exactly as before; only the name changed, and the phrase "boundary-value problem" now survives in this letter solely where Point 24 quotes the sentence that was withdrawn.

Point 32: The artifacts that were not in the transmitted files.

The reviewer notes that `scripts/spacing_convention_check.py` and its JSON, the externalized tables of the supplement, the figures and a compiled PDF were not part of the two files examined, so the 5×10^{-16} invariance and the build could not be checked. All of them are in the deposited repository, `https://github.com/Ahmedalgeria778/PoF_cavity_dissipation_study`: the script and `results/d2_derived/spacing_convention_check.json`, which contains the new ratio keys of Point 26; the six branch captures `capture_B1_D2_N256.npz` and its five siblings under `results/09_d2_capture/`; the table sources under `supplementary/data_appendix/`; the figures; and the four compiled PDFs, the two clean ones and the two marked ones. Both documents were recompiled after the changes above and the build logs are clean: no undefined references, no undefined citations, no duplicate labels, and every externalized table resolves.

Summary of Changes

Title: Changed from "Physics-informed identification and energetic characterization of conditional modal collapse in active control of lid-driven cavity flow" to "Spatio-temporal wall-actuation modes and cycle-mean viscous dissipation in a lid-driven cavity."
Abstract: Completely rewritten to lead with the physical question and announce the main quantitative results.
Introduction: Completely rewritten to present the physics of wall actuation, the energy budget, and the central question. The PINN is mentioned only as a tool for identifying branches.
Sections: Reorganized from 9 sections (focused on PINN identification) to 8 sections (focused on fluid physics); the symbolic-regression section of the previous version has been dropped.
Figures: Reduced from 17 figures to 8 figures. Six figures are carried over (old Figs. 10–11, 13–16), two are new, and the eleven dropped figures are the nine identification-campaign figures already published in Beniaiche (2026) plus the quasi-stationarity and symbolic-regression diagnostics.
Tables: Reduced from 11 tables to 4 tables in the main text. The detail tables — those already published in Beniaiche (2026) and those reporting the identification hyperparameters, the seed-level metrics and the grid-convergence data — are collected in the supplement, which now holds 19 tables (Tables S1–S19).
References: Reorganized to prioritize fluid physics over machine learning. Added Beniaiche (2026) as an antecedent. The bibliography grew from 25 to 37 entries, each checked against the publisher record before citing.
Supplementary material: Restructured into five sections (S1–S5): PINN implementation, control-space and modal-analysis details, identification-campaign robustness, the independent MRT-LBM verification with grid uncertainty, and the reproducibility inventory. It contains 19 tables, 2 figures and its own short bibliography.
Marked-up versions: Prepared for both the manuscript and the supplementary material using a paragraph-level diff script.
Highlight image: Prepared showing the strain-rate distribution for the three actuation branches.

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